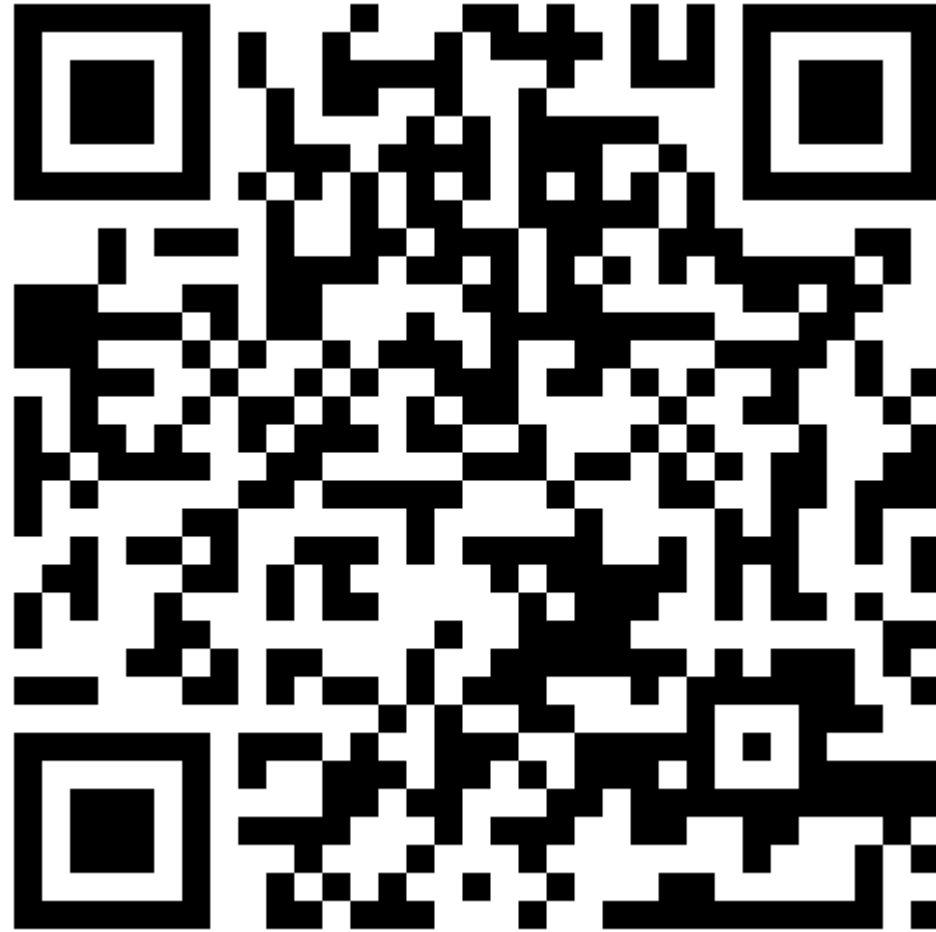


Attendance



<https://forms.gle/wwQiFmEqrrqXMDYcZ8>

Last time: Discrete-log assumption:

Definition: we say that DLOG is hard in a finite cyclic group $\mathbb{G}$ if for all PPT algorithm A ,

$$\Pr[A(\mathbb{G}, q, g, g^x) = x] \leq \text{negl}(n)$$

where g is a generator, q is the order of the group with $|q| = n$, and $x \in \mathbb{Z}_q$ is chosen uniformly

Group

Group: under 1 operation $\circ$

1. **Closure**: For all a, b in G , the result of the operation, $a \circ b$, is also in G
2. **Associativity**: For all a, b and c in G , $(a \circ b) \circ c = a \circ (b \circ c)$
3. **Identity element**: There exists an element e in G such that, for every element a in G , the equation $e \circ a = a \circ e = a$ holds. Such an element is unique
4. **Inverse element**: For each a in G , there exists an element b in G , such that $a \circ b = b \circ a = e$, where e is the identity element

Commutativity: If for all a, b in G , $a \circ b = b \circ a$, the group is called **abelian**

Examples: $\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\}$ with $+$ mod N

Notation

- N denotes a positive integer
- p denotes a prime
- $\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\}$
- Modular arithmetic in $\mathbb{Z}_N$ $+$ and $\times \bmod N$

Simplifying modular arithmetic

- $a + b + c \bmod N = (a + b) \bmod N + c \bmod N$
- $a \times b \times c \bmod N = (a \times b) \bmod N \times c \bmod N$

Defining multiplicative group $\mathbb{Z}_N^*$

Def: $\mathbb{Z}_N^* =$ (set of invertible elements in $\mathbb{Z}_N$)
 $= \{ x \in \mathbb{Z}_N : \gcd(x, N) = 1 \}$

Lemma: $x \in \mathbb{Z}_N$ has an inverse if and only if $\gcd(x, N) = 1$

GCD: Euclidean algorithm

$a \cdot x + b \cdot y = \gcd(x, y)$, extended Euclidean algorithm

https://en.wikipedia.org/wiki/Extended_Euclidean_algorithm

Cyclic group and generator

Definition: $\mathbb{Z}_p^*$ is a cyclic group, if

$$\exists g \in \mathbb{Z}_p^* \text{ such that } \{1, g, g^2, \dots, g^{p-2}\} = \mathbb{Z}_p^*$$

Example $p = 7$: $\{1, 3, 3^2, 3^3, 3^4, 3^5\} = \{1, 3, 2, 6, 4, 5\} = \mathbb{Z}_7^*$

Definition: $\mathbb{G}$ is a cyclic group, if

$$\exists g \in \mathbb{G} \text{ such that } \{e, g, g^2, \dots, g^{|\mathbb{G}|-1}\} = \mathbb{G} \text{ by applying the group operation on } g$$

Discrete-log assumption:

Given $y = g^x$, hard to compute x in some groups

Definition: we say that DLOG is hard in a finite cyclic group $\mathbb{G}$ if for all PPT algorithm A ,

$$\Pr[A(\mathbb{G}, q, g, g^x) = x] \leq \text{negl}(n)$$

where g is a generator, q is the order of the group with $|q| = n$, and $x \in \mathbb{Z}_q$ is chosen uniformly

Diffie-Hellman key exchange

- Proposed by Whitfield Diffie & Martin Hellman in 1976
 - Different way of doing crypto than approaches in the previous 4,000+ years
 - Foundation for public key crypto (RSA, ElGamal, etc)



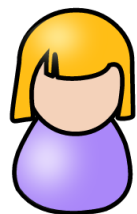
- Side notes:
 - Merkle credited by Hellman as a strong inspiration for the design
 - Similar method developed in the 1960s at GCHQ (UK) by James Ellis, but classified...

Diffie-Hellman key exchange

Public information: $\mathbb{Z}_p^*$ defined by a large prime p and generator g

Choose random $x \in \mathbb{Z}_p^*$

Alice



$$X = g^x \bmod p$$



$$Y = g^y \bmod p$$



Choose random $y \in \mathbb{Z}_p^*$

Bob



$$g^{xy} = (Y)^x \bmod p$$

$$g^{xy} = (X)^y \bmod p$$

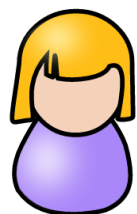
The goal is to generate a common secret!

Diffie-Hellman key exchange

Public information: $\mathbb{Z}_p^*$ defined by a large prime p and generator g

Choose random $x \in \mathbb{Z}_p^*$

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Bob



$$g^{xy} = (Y)^x \bmod p$$

$$g^{xy} = (X)^y \bmod p$$

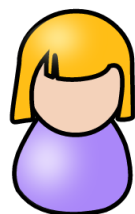
The goal is to generate a common secret!

Diffie-Hellman key exchange

Public information: $\mathbb{Z}_p^*$ defined by a large prime p and generator g

Choose random $x \in \mathbb{Z}_p^*$

Alice



Eavesdropper can see it!

$$X = g^x \bmod p$$



$$Y = g^y \bmod p$$



Choose random $y \in \mathbb{Z}_p^*$

Bob



$$g^{xy} = (Y)^x \bmod p$$

$$g^{xy} = (X)^y \bmod p$$

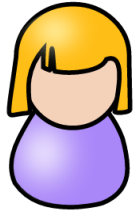
The goal is to generate a common secret!

Example

Public information: $\mathbb{Z}_{11}^*$, generator $g = 2$

Choose random $4 \in \mathbb{Z}_p^*$

Alice



$$X = 2^4 \bmod 11 = 5$$



$$Y = 2^5 \bmod 11 = 10$$



$$(Y)^x = 10^4 = 1 \bmod 11$$

Choose random $5 \in \mathbb{Z}_p^*$

Bob



$$(X)^y = 5^5 = 1 \bmod 11$$

Security of Diffie-Hellman

- Attacker: $\mathbb{Z}_p^*, g, X = g^x, Y = g^y$
- Discrete-log assumption: cannot compute x and y

Security of Diffie-Hellman

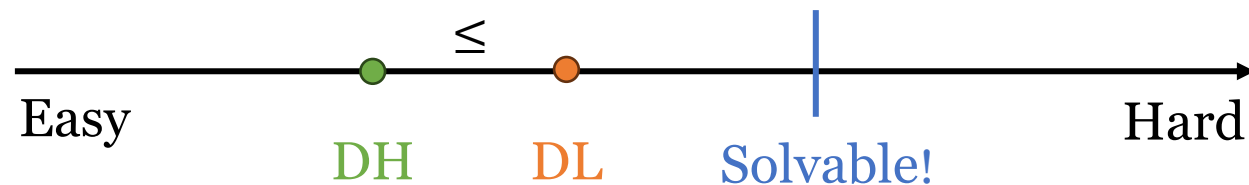
- Attacker: $\mathbb{Z}_p^*, g, X = g^x, Y = g^y$
- Diffie-Hellman assumption: given $\mathbb{Z}_p^*, g, g^x, g^y$, cannot compute g^{xy}

Relationship of problems and assumptions

- Discrete logarithm problem: given g^x , compute x
- Diffie-Hellman problem: given g^x, g^y , compute g^{xy}

Which one is easier and which one is harder?

- Diffie-Hellman problem is easier than discrete-log
- Problem A is easier than problem B, if given the solution of B, A can be solved in polynomial time.



Relationship of problems and assumptions

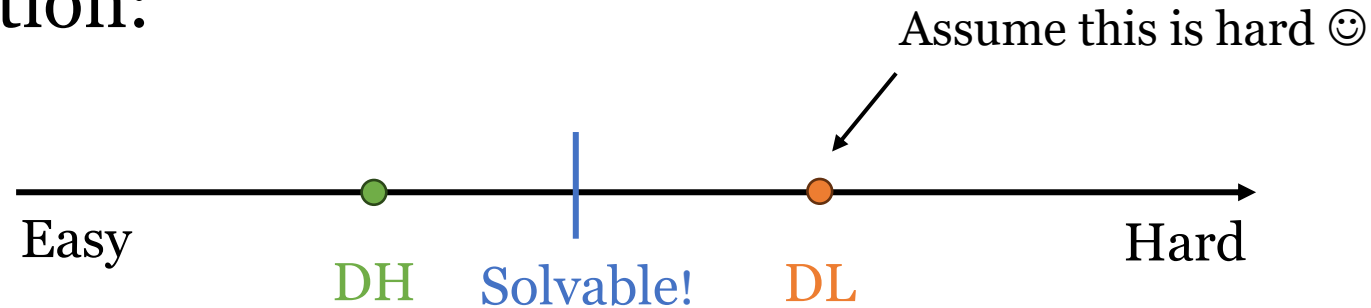
- Discrete logarithm assumption: Discrete-log problem is hard
- Diffie-Hellman assumption : Diffie-Hellman problem is hard

Which assumption is stronger and which one is weaker?

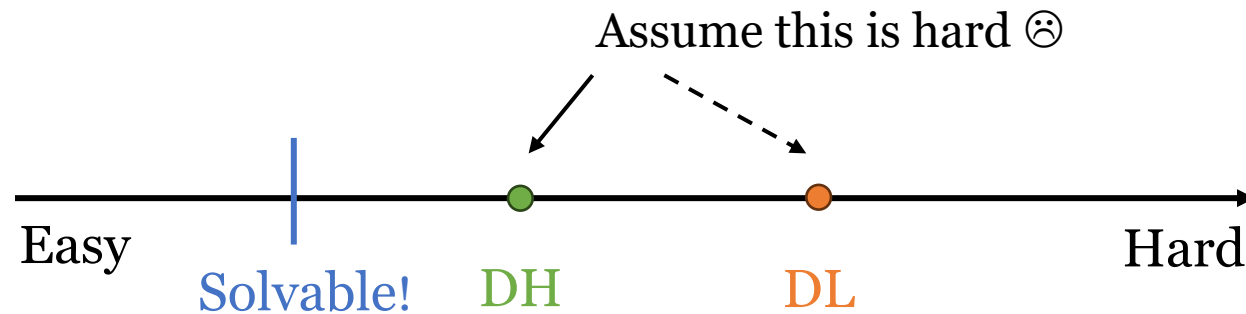
- Diffie-Hellman assumption is stronger than discrete-log
- Assumption A is stronger than assumption B, if problem A is easier than problem B

Weak assumption vs strong assumption

- Weak assumption:



- Strong assumption:

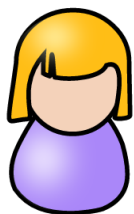


Diffie-Hellman key exchange on $\mathbb{G}$

Public information: $\mathbb{G}$ with order q and generator g

Choose random $x \in \mathbb{Z}_q$

Alice



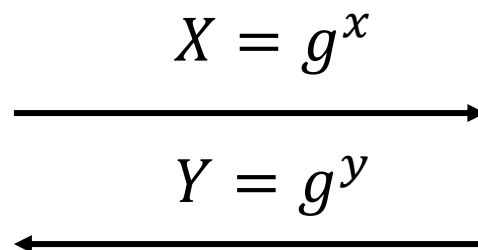
$$g^{xy} = (Y)^x$$

Choose random $y \in \mathbb{Z}_q$

Bob



$$g^{xy} = (X)^y$$



Computational Diffie-Hellman (CDH)

Assumption: we say that CDH is hard in group $\mathbb{G}$ if for all PPT algorithm A , there is a negligible function that

$$\Pr[A(\mathbb{G}, q, g^x, g^y) = g^{xy}] \leq \text{negl}(n)$$

where g is a generator, q is the order of the group, $x, y \in \mathbb{Z}_q$ are chosen uniformly.

Decisional Diffie-Hellman (DDH) Assumption

Assumption: we say that DDH is hard in group $\mathbb{G}$ if for all PPT algorithm A , there is a negligible function that

$$\Pr[A(\mathbb{G}, q, g^x, g^y, g^z) = 1] - \Pr[A(\mathbb{G}, q, g^x, g^y, g^{xy}) = 1] \leq \text{negl}(n)$$

where g is a generator, q is the order of the group, $x, y, z \in \mathbb{Z}_q$ are chosen uniformly.

Relationships

Assumption:

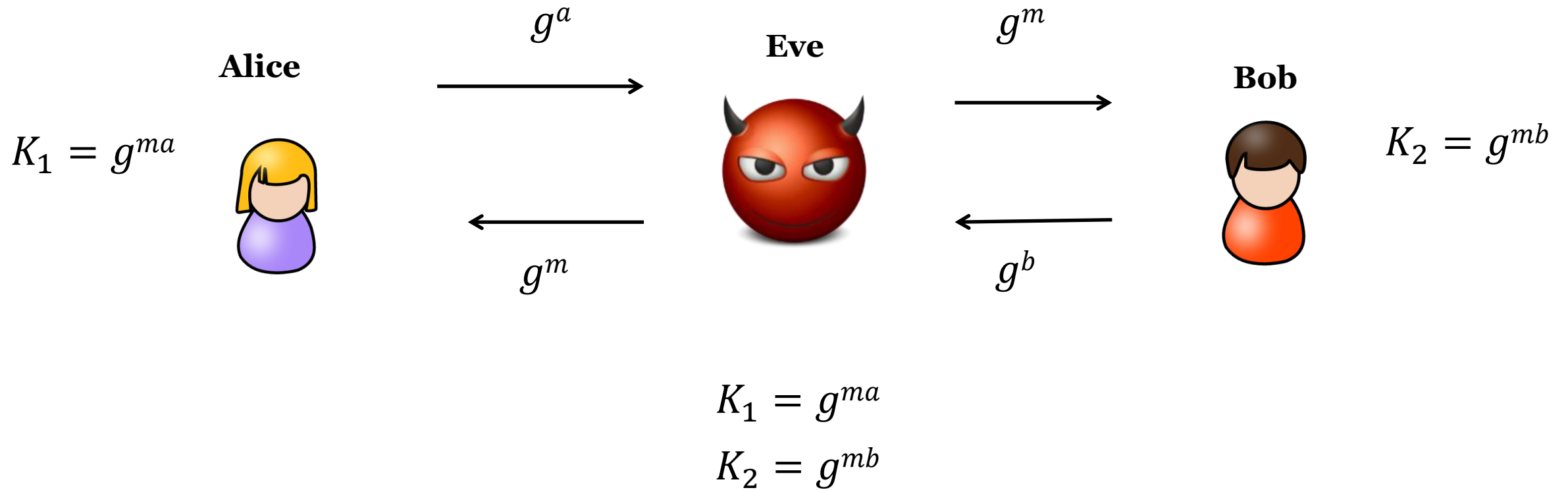
DDH $\rightarrow$ CDH $\rightarrow$ DLOG



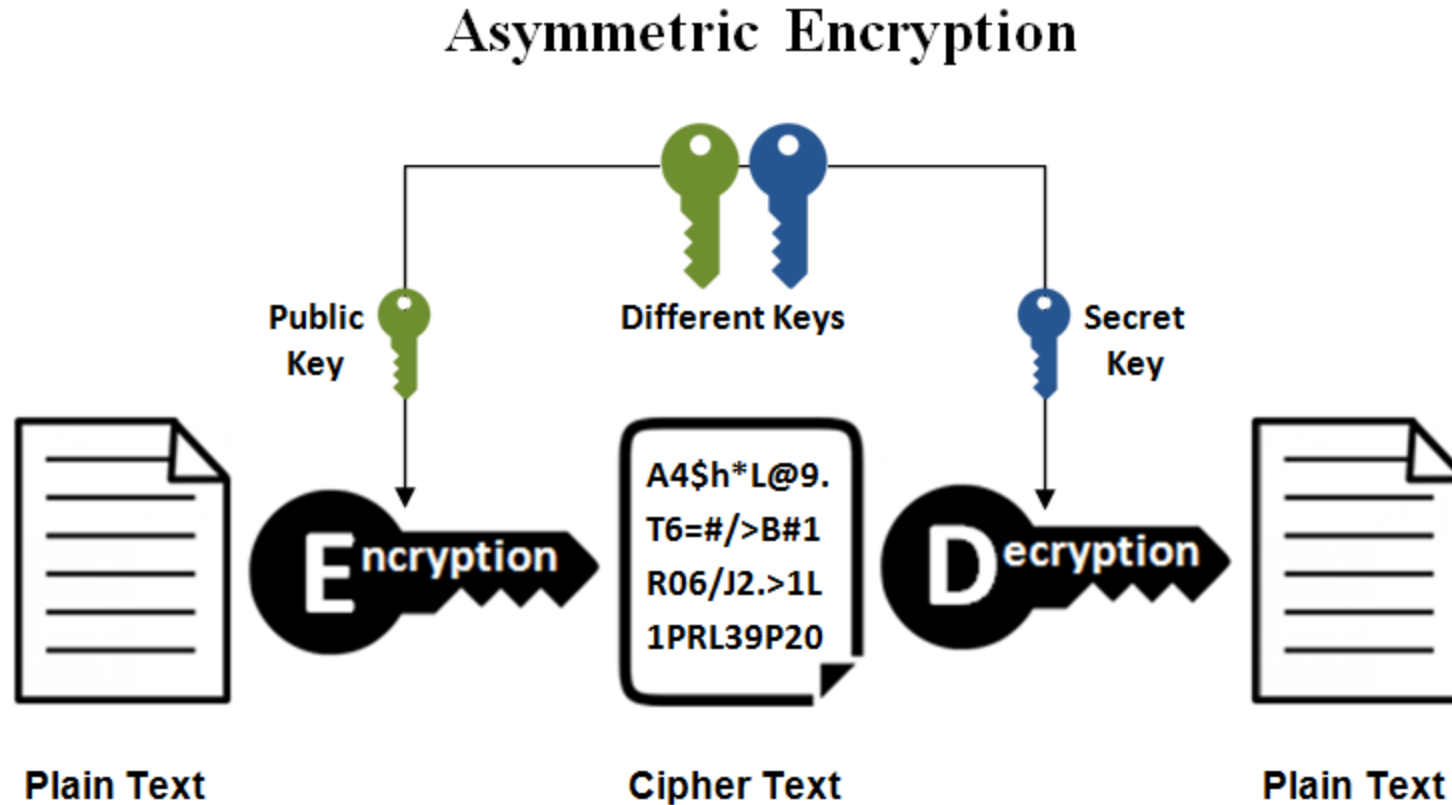
Stronger

Weaker

Man-in-the-Middle attack



Public key encryption (Asymmetric)



Definition of public key encryption

- Algorithms:

- sk : private key, pk : public key, m : message, c : ciphertext
- $(pk, sk) = \text{Gen}(1^n)$
- $c = \text{Enc}(pk, m)$
- $m = \text{Dec}(sk, c)$

Correctness: $m = \text{Dec}(sk, \text{Enc}(pk, m))$ for $(pk, sk) \leftarrow \text{Gen}(1^n)$

Distribution of public keys

- We assume that the sender can obtain the PK via an authenticated channel
- Public-key infrastructure (PKI)

Chosen-plaintext attack (CPA) security

Experiment $\text{PubK}_{A,\Pi}^{\text{CPA}}(n)$:

1. (pk, sk) is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n , and **oracle access to $\text{Enc}_{pk}(\cdot)$** . A outputs two messages $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1|$
3. A random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_{pk}(m_b)$ is given to A .
4. A continues to have **oracle access to $\text{Enc}_{pk}(\cdot)$** and outputs b'
5. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Definition: a public key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

$$\Pr[\text{PubK}_{A,\Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

Security against eavesdropper

Experiment $\text{PubK}_{A,\Pi}^{\text{EAV}}(n)$:

1. (pk, sk) is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n and pk . A outputs two messages $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1|$
3. A random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_{pk}(m_b)$ is given to A .
4. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Theorem: if a public-key encryption scheme is EAV secure, it is CPA secure.

Multiple encryptions

- Similar to symmetric key encryption, if a public key encryption is CPA secure, it is secure for multiple encryptions.

El Gamal Encryption

- Gen: let $\mathbb{G}$ be a finite cyclic group of order q and generator g .
Pick $x \in \mathbb{Z}_q$ randomly and set $h = g^x$.

The public key is $pk = (\mathbb{G}, q, g, h)$ and the private key is $sk = x$.

- Enc: on a message $m \in \mathbb{G}$, choose a random $r \in \mathbb{Z}_q$ and output

$$c = (g^r, h^r \cdot m)$$

- Dec: on a ciphertext $c = (c_1, c_2)$, compute

$$m = c_2 / c_1^x$$

Correctness

- If $c = (c_1, c_2) = (g^y, h^r \cdot m)$, then $\frac{c_2}{c_1^x} = \frac{(g^x)^r \cdot m}{(g^r)^x} = m$